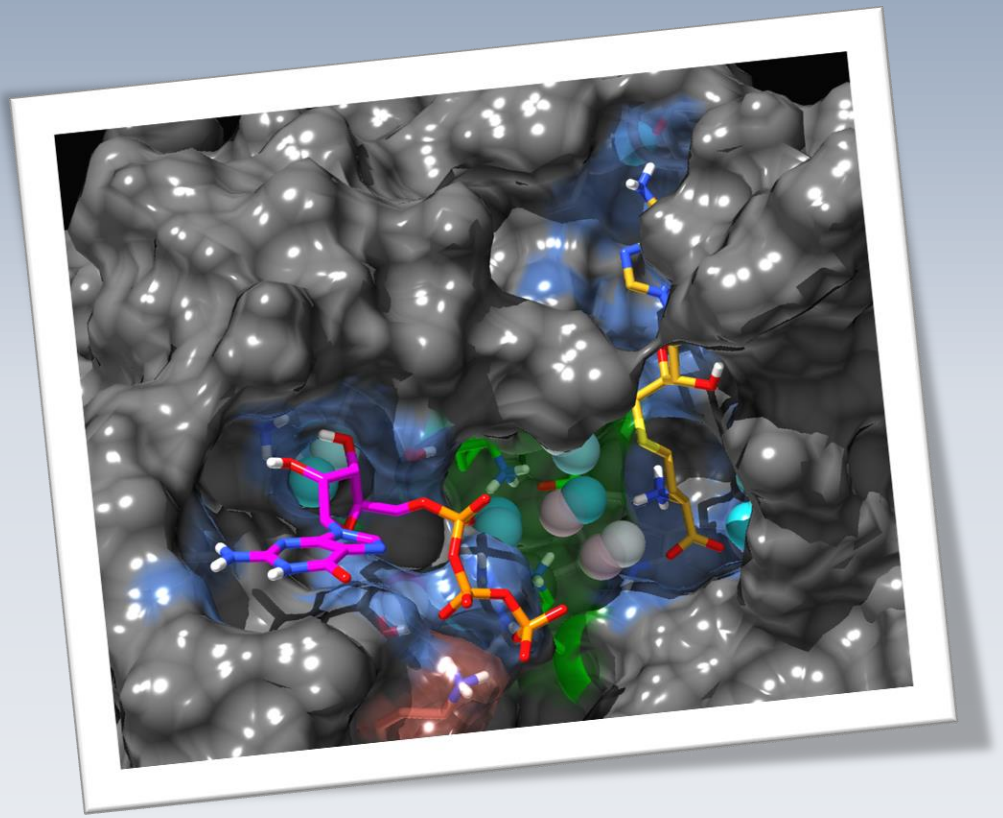


A Brief Introduction to Molecular Modeling

[1] Basic Application of Molecular Graphics

[2] Basic Principles and Applications of Protein-ligand docking

(October 2020 v1, August 2021 v2)
(September 2021 merged v1)

Table of Contents

1. Preface
   i. Expectation vs Reality
   ii. Research planning
2. Biochemistry recap
3. Preparation
   i. Importance of literature review
   ii. Identify the purpose of docking
   iii. Selection of receptor
   iv. Selection of binding site
   v. Selection of the ligands
4. Basic concepts of docking
4. Practical example
   i. Recommended software
   ii. Method 1: Chimera-Vina
   iii. Method 2a: Chimera-ADT-Vina
   iv. Method 2b: Chimera-ADT-AD4
   v. Using the command line
5. Reference

Preface

Expectation vs Reality

Research Planning

Your expectations?

1. What kind of “research experience” you are looking for?
2. What you expect a computer can do?

Reality

1. What kind of “research experience” you are looking for?
   - Research =/= achievement
   - Academia =/= let you hide from the job market
2. What you expect a computer can do?
   - Main advantages:
     - Virtually screen and design inhibitors → reduce experimental cost
     - Study phenomena that are difficult to be investigate experimentally
   - You think its easier to handle than experimental work?
     - NO! the learning curve can be much steeper
   - **LIMITS: Computer cannot do everything**
     - →**Still have false positive/false negative hits!!!!!**
     - →**Note the assumptions you are making!!!!!!**
     - →**Always try to validate your results if possible!!!!!**
     - →**Be VERY careful with the predicted/generated results!!!!!!**

What does it mean by doing research?

1. Identify the problem to be investigated
2. Find the possible methods to solve the identified problem
   a. Identify the advantages and limits of each methods
   b. Is it the only solution?
3. Identify the resources you can afford for this problem
   a. Time to learn the method
   b. Materials/instruments/computational power you have
   c. Using the current resources, estimate the time you need to do the task
4. Choose the appropriate method(s) and **plan** for your work
   a. "write your own manual" for you own topic
   b. Split your work into a few levels:
      i. What MUST be finished in order to survive (minimal requirements)
      ii. What can be done after meeting the minimal requirements, to improve your work
      iii. Bonus stuff that make your work looks fancy but maybe very difficult to do
   c. Prepare a "plan B" in case that you can't fulfil the minimal requirements with the original method
5. ~~Do~~ Try it. If failed: find the reason, adjust your plan and continue.

Preparation

- Prepare well before you start!
- Drafting a research plan yourself is highly recommended.
  - Brief examples will be given in the next few slides for you to quickly try it first.

Planning your (computational) experiment

- Resources on hand
- Methods you choose
- List of tasks to be finished
  - Answer you can get by finishing the task
  - Preparation you need to do before doing the task
  - How to do the task
  - How to analyze your results
  - Priority of different tasks
- A rough schedule with the estimated time to finish each task

➢ The plan may change after you start working on it, but drafting it at the beginning helps to clear your mind, get the "big picture" of your project and ensure you have something minimal when you reach a deadline

Note the resources you can afford

- Time:
  - Project deadline
  - To learn the technique
  - To apply the skills you have learnt
  - To spend on trial-and-error and optimizing the experimental setup
  - To run the simulation
  - To analyze and interpret the results
- Hardware:
  - Better hardware → speed up
  - For MD: many codes utilise the GPU to accelerate the calculation
- Software:
  - Compatible with your computer?
  - Compatible with your OS?
  - Able to install?

Setting the research problem

- Why is your research important?
- Any questions still need an answer on this topic?
- What answer you want to get from doing research?

Analysis to perform

- What you need to analyze from the simulation output?
- How to interpret the raw data you get to make it something meaningful? Something that can answer your research problem?
- How to validate your results?
- Are your results reproducible?

Using the case of ZIKV MTase…

- What you want to see from docking/MD simulation?
  - Enzyme-substrate interaction?
  - Design a new active-site inhibitor? Design an allosteric regulator?

→This would affect:

- Simulation method
- Simulation timescale
- System to be simulated
- Analysis to be done

Case of ZIKV MTase

1. Purpose of the study?
   - Binding affinity?
   - Binding pathway? May need different tools for a quick search/special MD techniques
   - Conformational change upon substrate binding? Use MD instead of docking
2. Protein structure to use?
   - MTase domain only?
   - Full-length NS5?
   - Comparative study with human MTase?
3. Time and computational resources you can afford for this project?
4. Outline what you want to and how to do it!
   - Research plan
   - Research outline

Prerequisite

Basic biochemistry you need to know for making molecular graphics or to perform docking analysis

Protein Structure

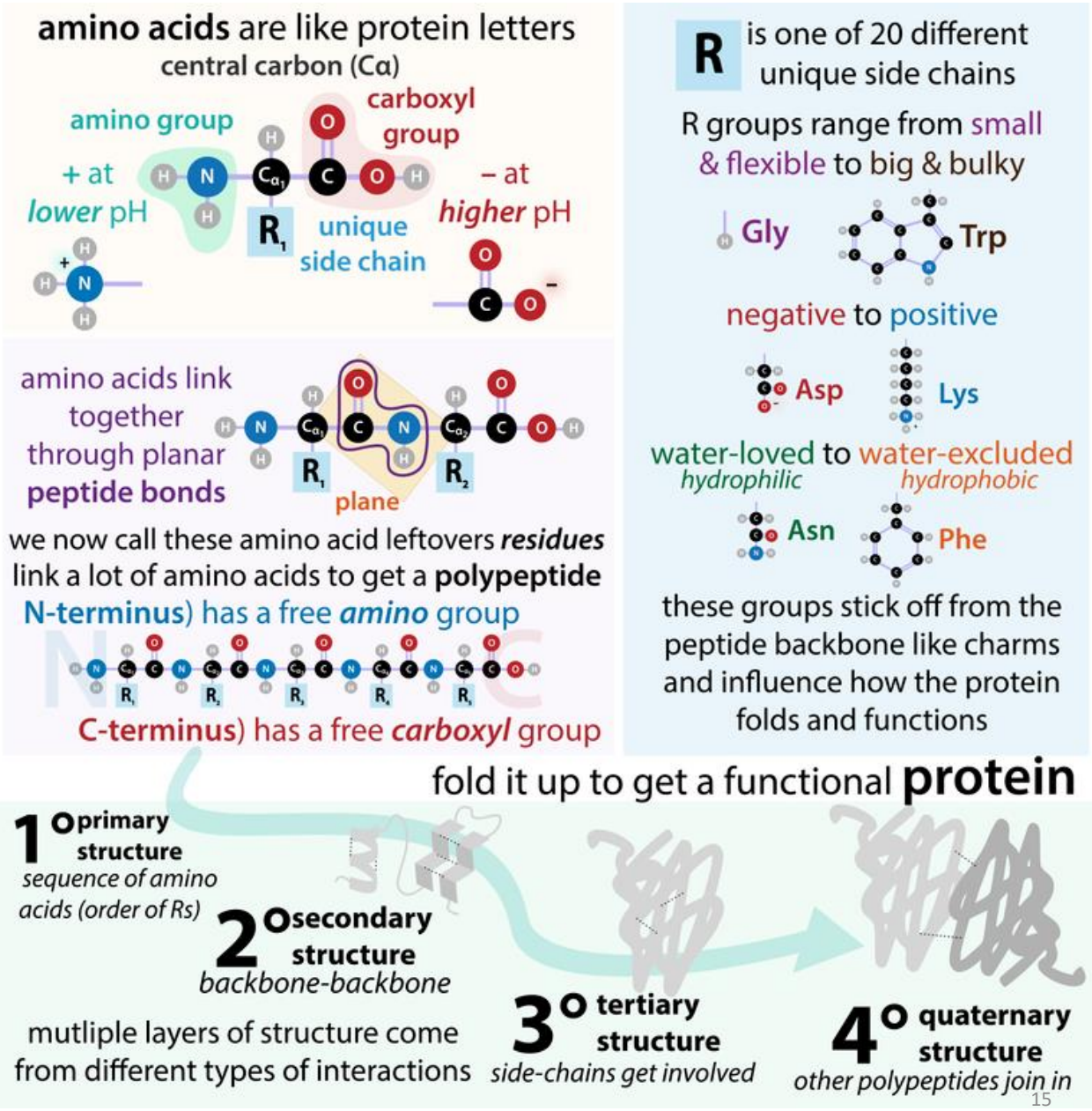


Image adapted from wikimedia commons

20 Amino Acids

Structure

Name

(3-letter code / 1-letter code)

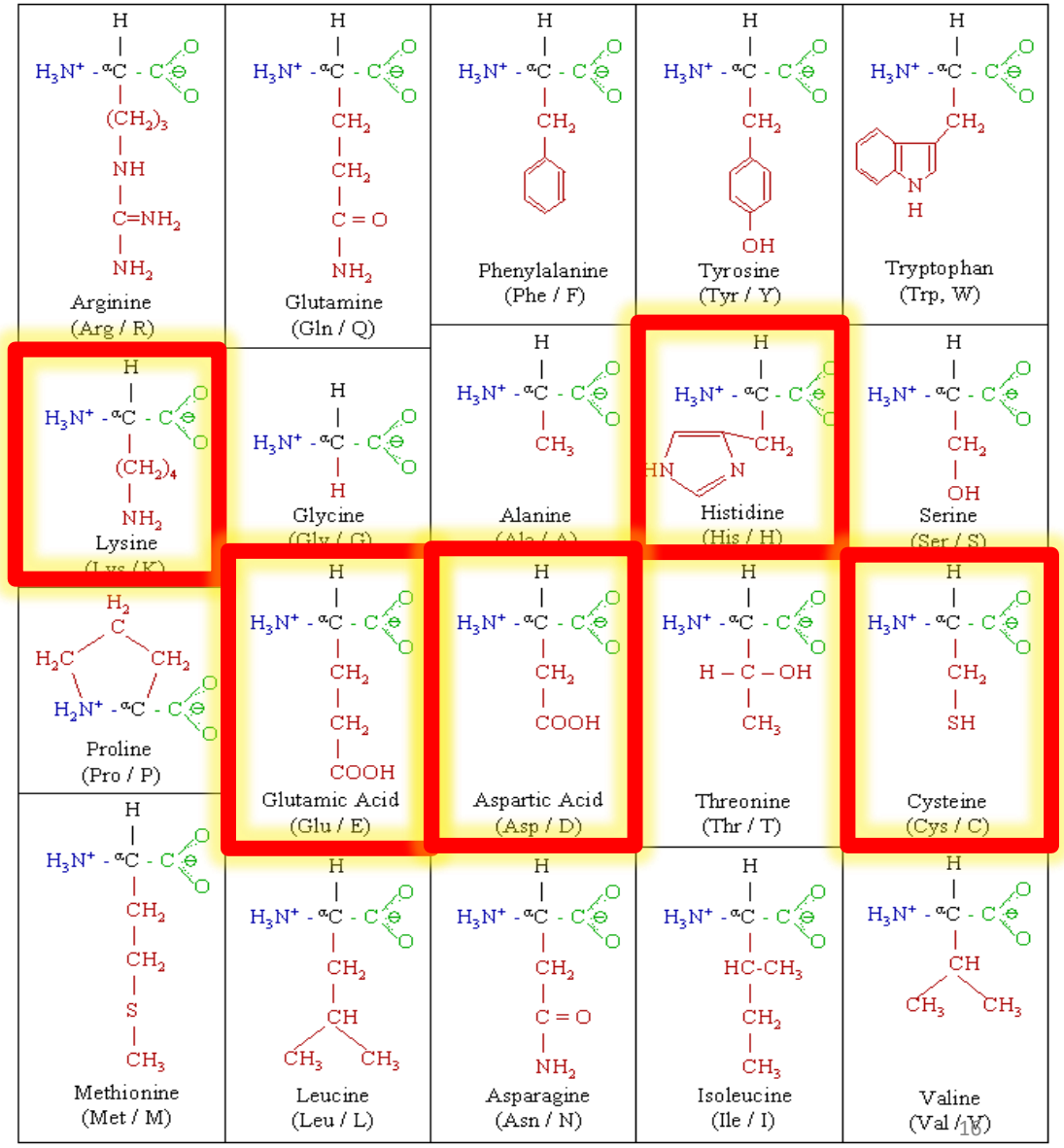
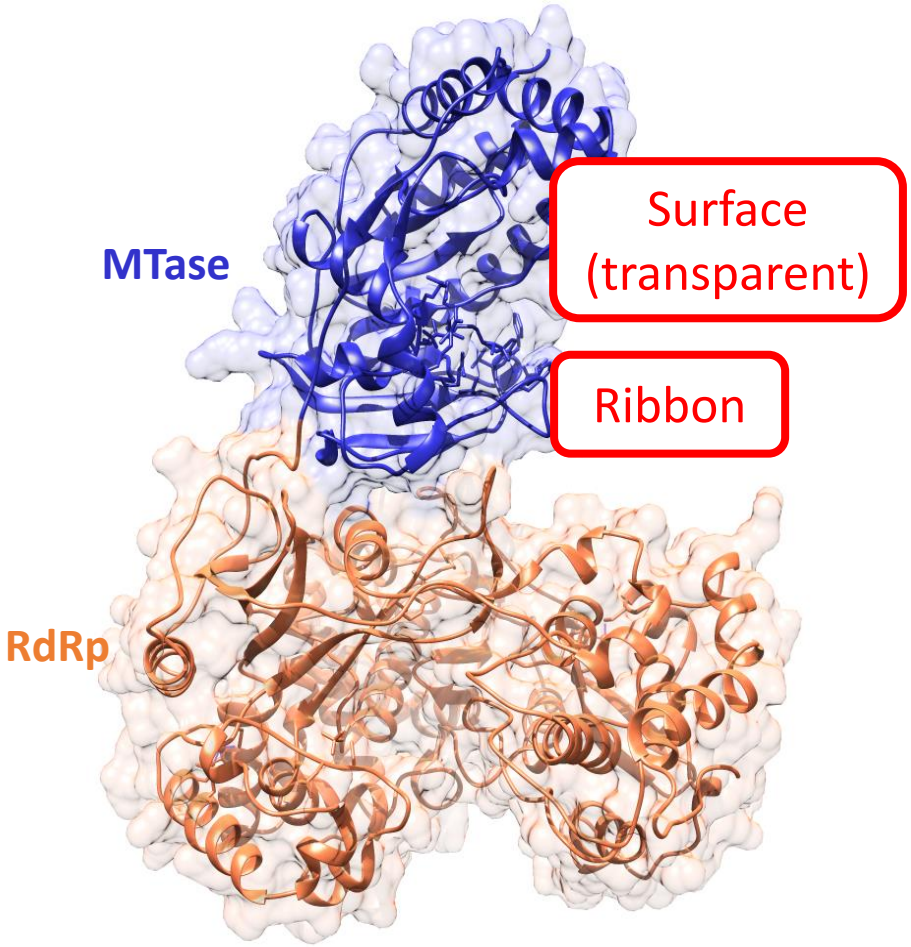


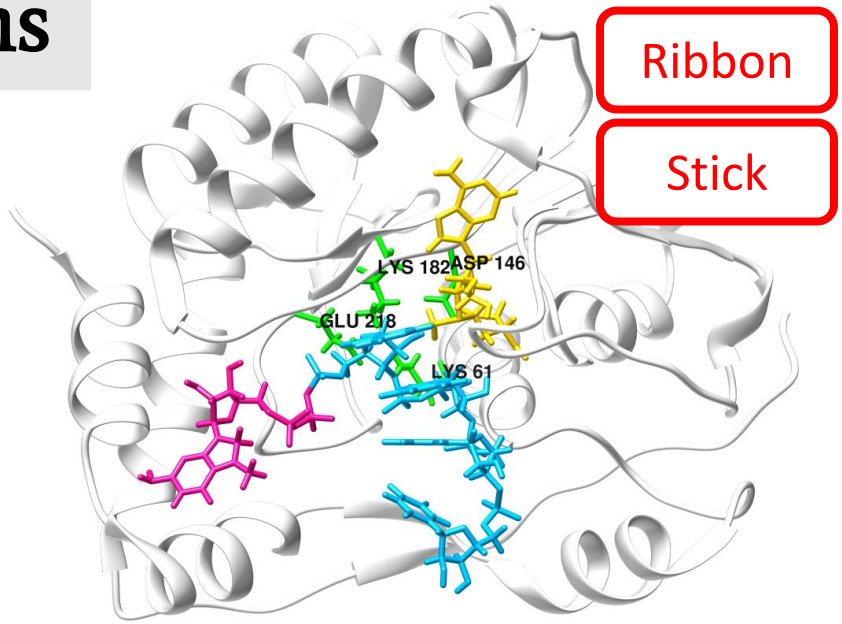
Image adapted from wikimedia commons

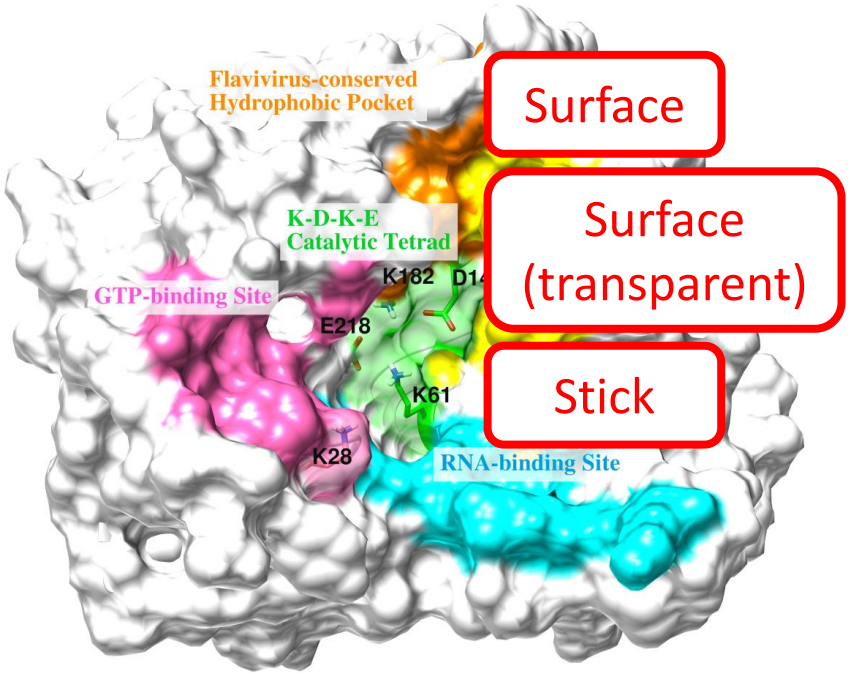
Representation of Proteins

Images generated with UCSF Chimera.

PDB: 5ULP, 5MRK, 5DTO, 5KQS





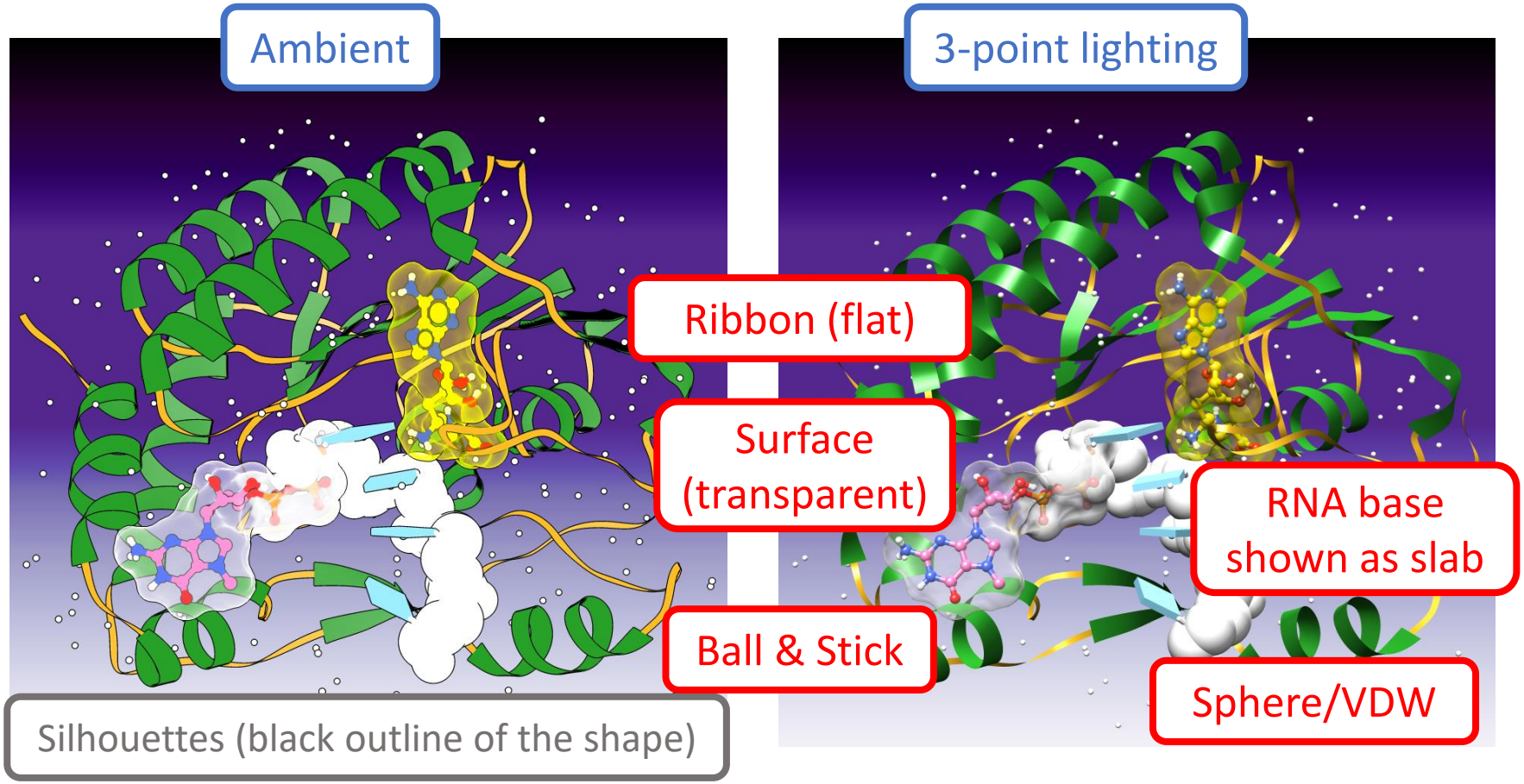


Representation of Proteins

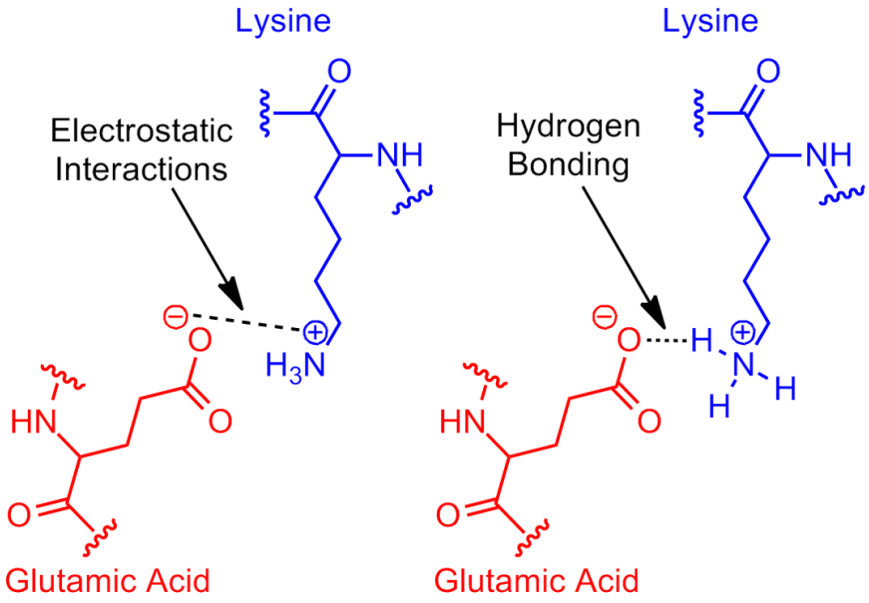
Christmas cards I made last year.

Images generated with UCSF Chimera.

PDB: 5ULP, 5MRK, 5DTO, 5KQS



Molecular Interactions

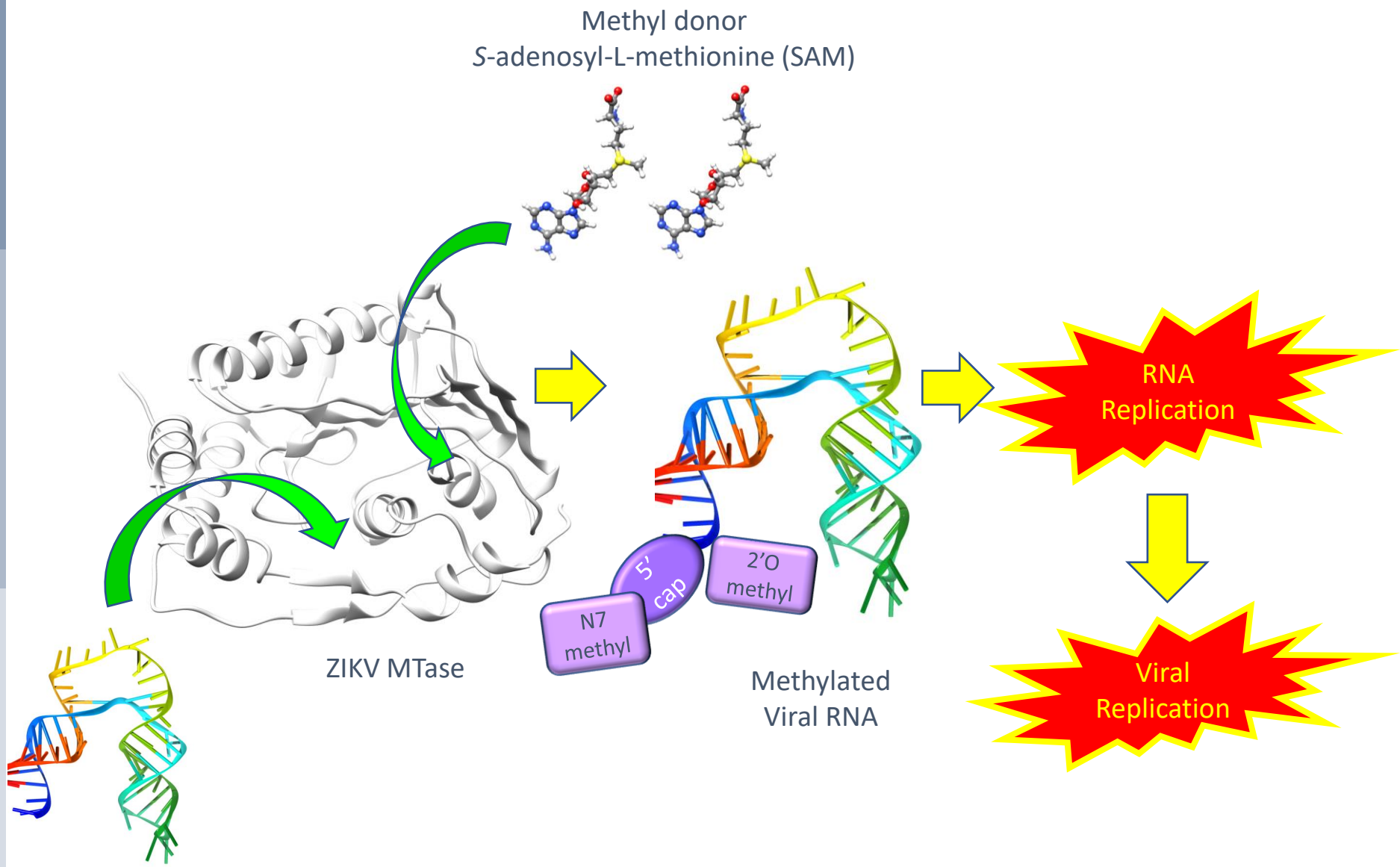


Source: https://i.stack.imgur.com/Xmnln.png

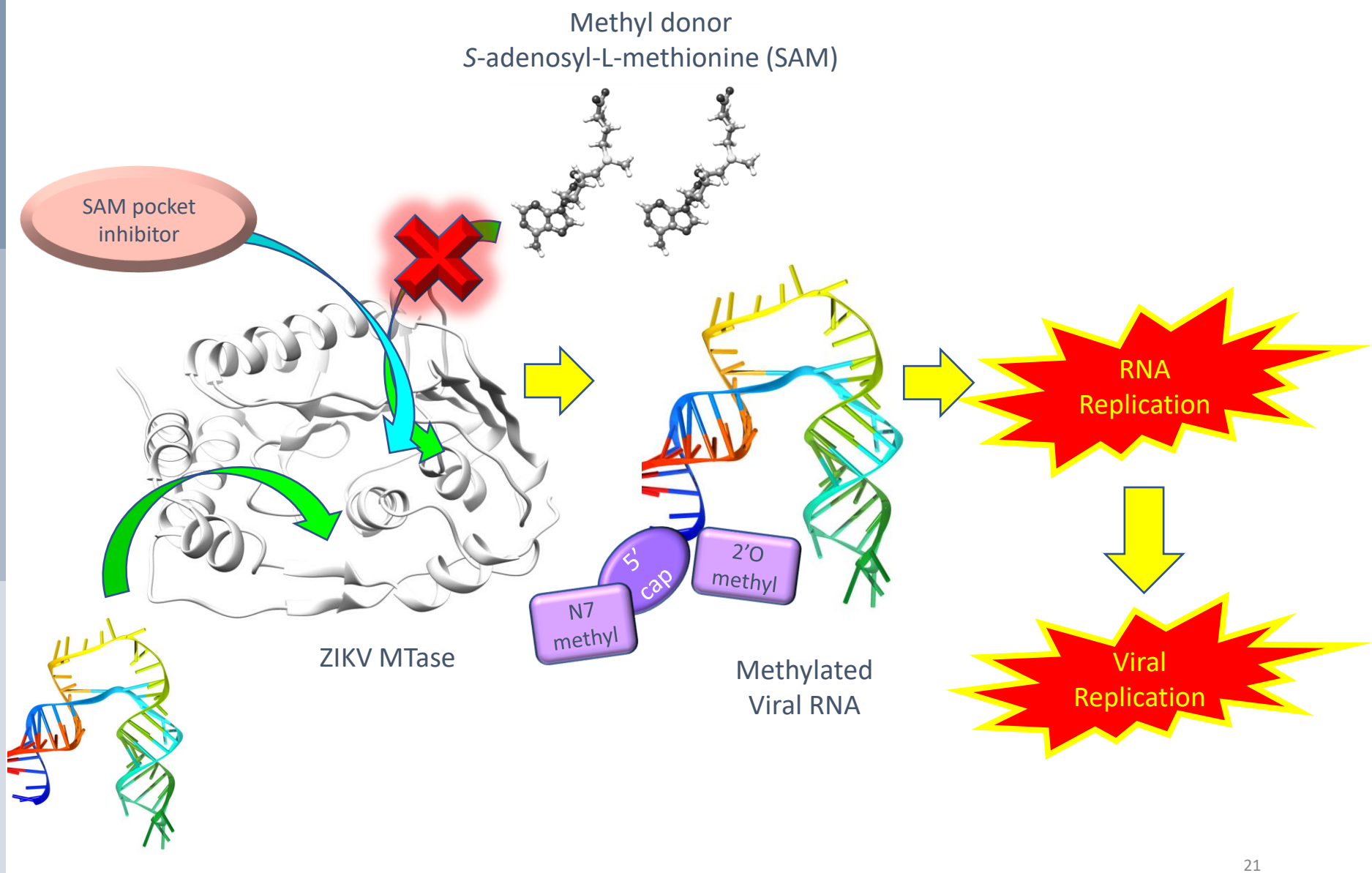
Also:

- VDW interactions
- Pi-interactions
- …

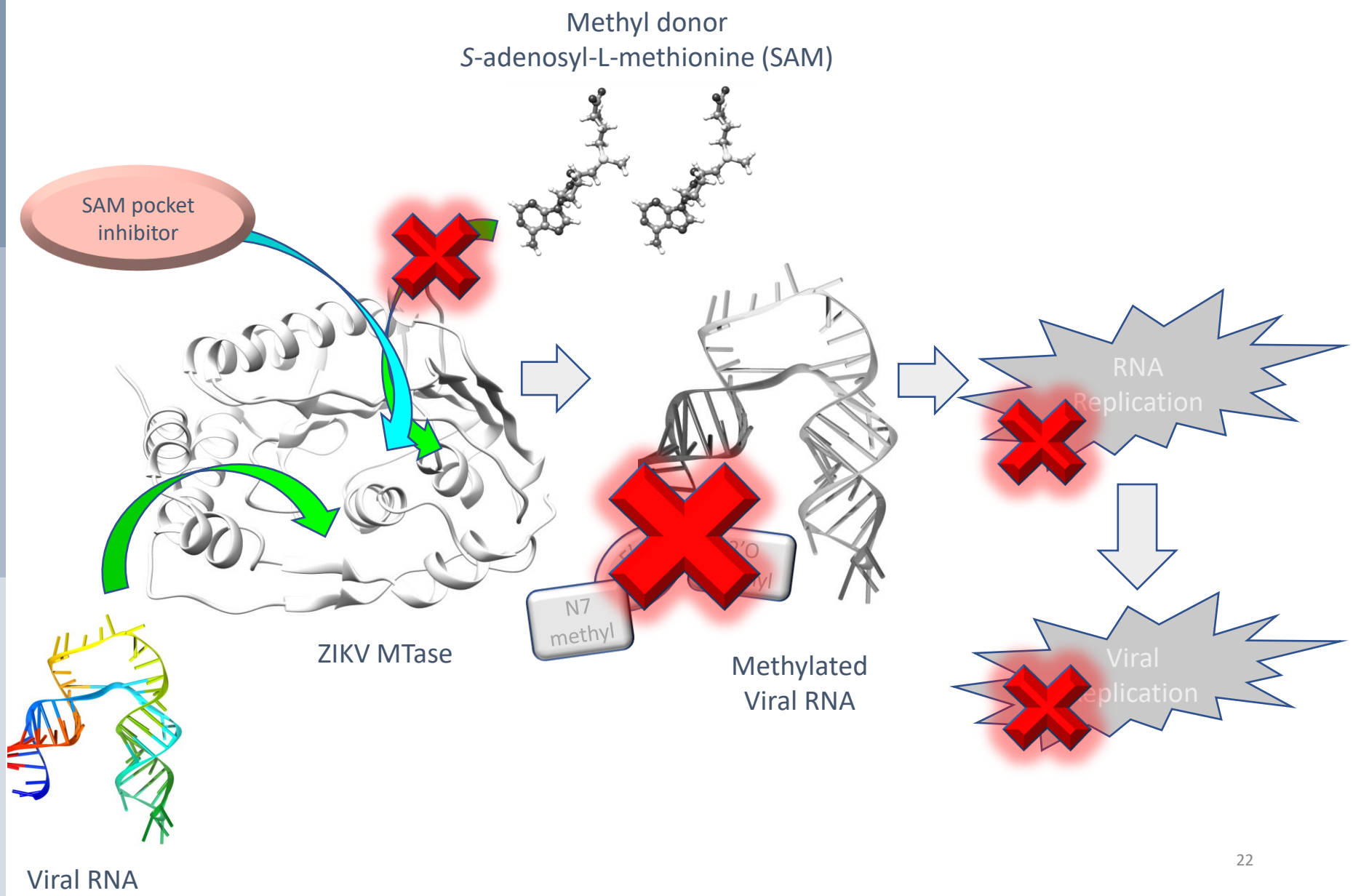
Function of an enzyme catalyzing certain reaction or biological process



Inhibition of an enzyme terminate enzymatic actions → alter subsequent processes



Inhibition of an enzyme terminate enzymatic actions → alter subsequent processes



Preparation

What you need to do before you perform docking

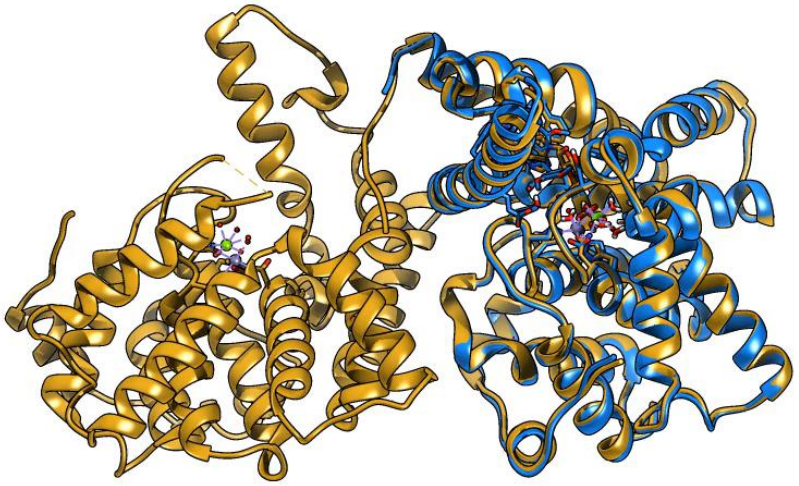
Importance of literature review

- Function of enzyme
- Catalytic mechanism
- Mutations
- pH of environment:
  - Physiological pH?
  - Acidic (e.g. stomach)?
- Protonation especially around the binding site
- Metal coordination
- Ions
- Water
- Membrane protein?

Importance of literature review

- Example: PDE10A
  - Complicated environment: be careful with the step of input preparation!
  - Need to refer to multiple literature (if available) to have a model relatively accurate to reproduce experimental observations

Phosphodiesterase 10A (PDE10A)



Identify the purpose of docking

- MD input preparation of unknown molecule in known receptor
- Virtual screening of novel inhibitors
- Brief insights of Inhibition mechanism
- Quick characterization of protein-ligand interactions
- Study the effect of mutation
- Designing covalent probe of specific residue
- QSAR
- …

Selection of receptor starting structure

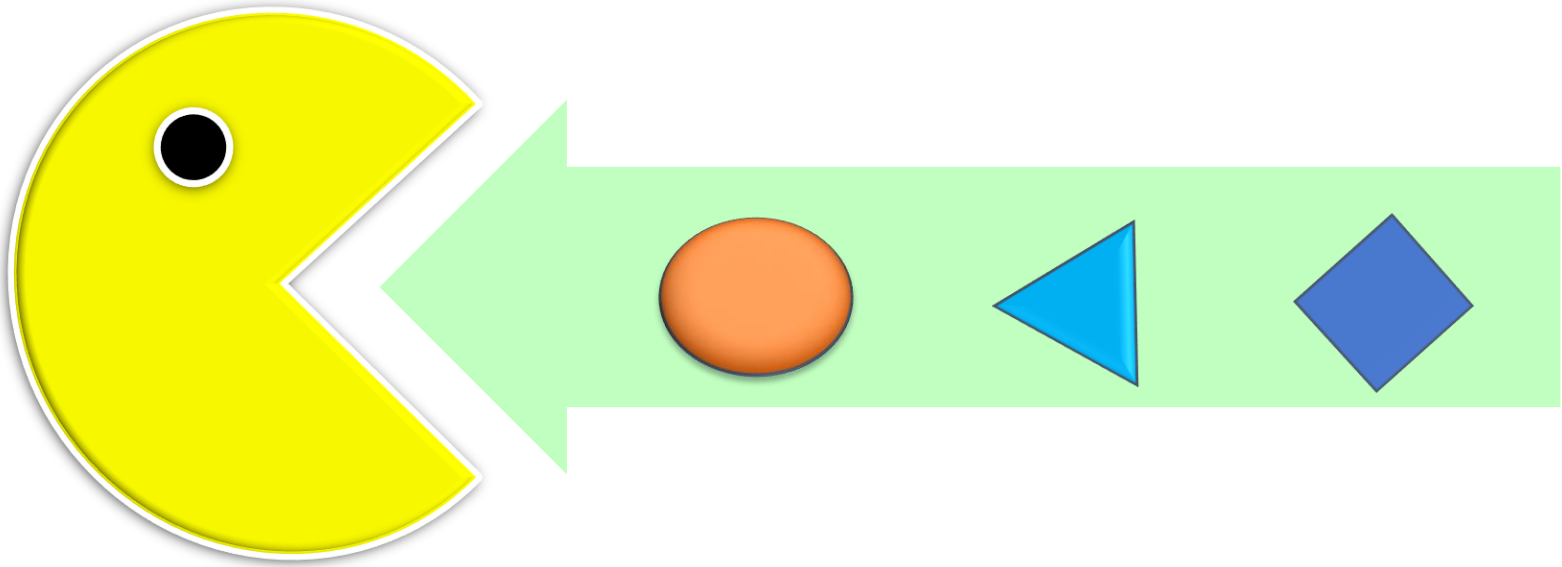
- Apo (unliganded) vs ligand bonded
- Complexed metal ions
- mutations?
- resolution

Selection of binding site to be targeted

- Approaches
  - choosing a known pocket
  - pocket detection
  - blind docking
  - notes on allosteric site: choice of structures

Selection of the ligands to be modelled

- Co-crystallized ligand
- Active compounds from literature without a enzyme-bounded crystal structure
- Ligands from a similarity/pharmacophore search
- Virtual screening of a molecule database
- *De novo* generated compounds using different methods
- Ligands that you design yourself
- …



Basic concepts of docking

A very brief introduction to the theories behind molecular docking

General

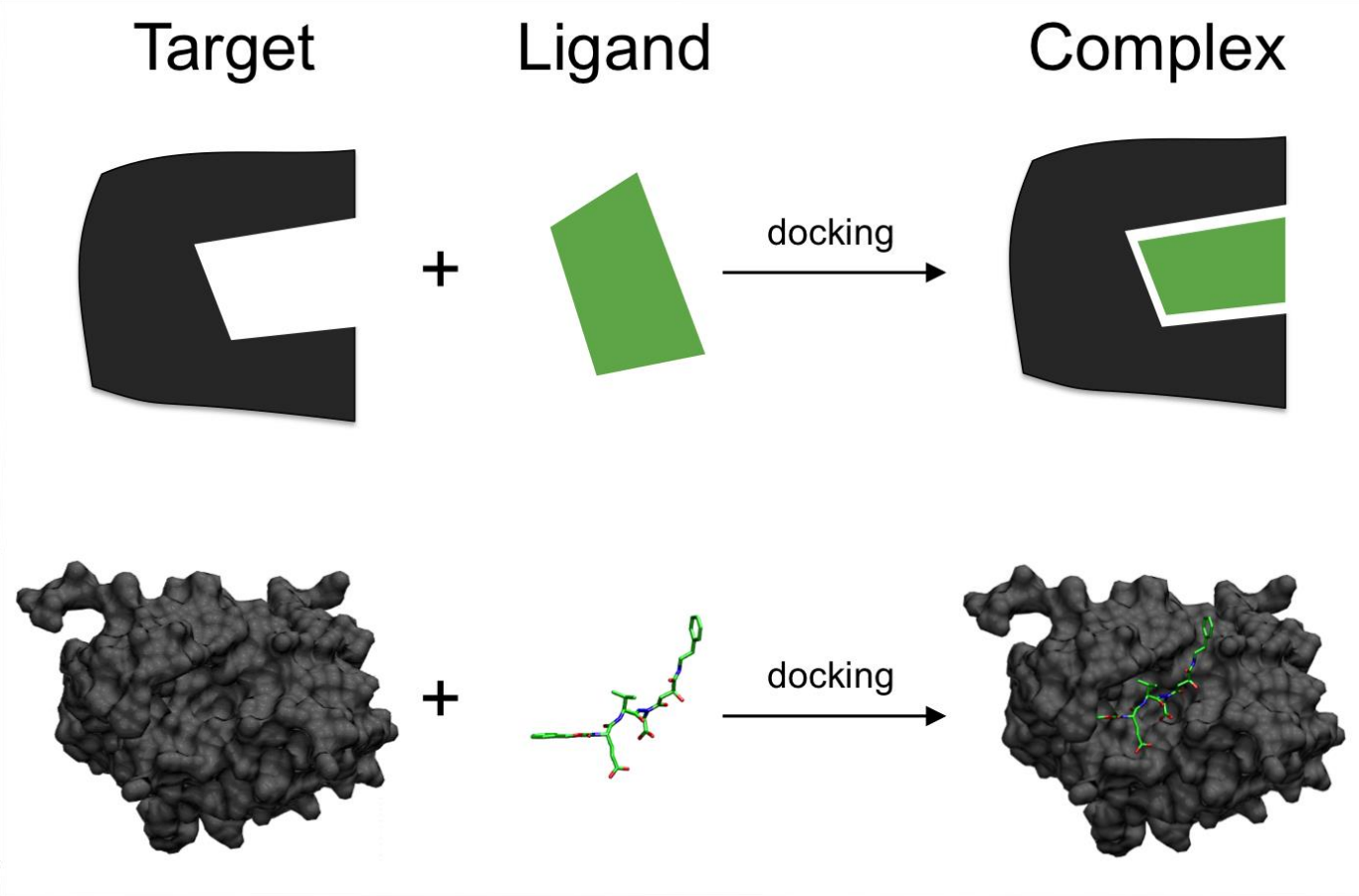


Image adapted from Wikimedia commons

Conformational search

- Conformational search v.s. geometry optimization
- Methods of conformational search
  - Systematic vs random search
  - Simulation based method
  - Genetic algorithm

Scoring methods

- Examples
  - Force-field based
  - Alignment based/pharmacophore based
  - Based on a QSAR model
  - Combination of methods
- In many cases the scores are relative to system and many have limited transferability

Rigid docking, flexible docking and MD

- Speed vs accuracy
  - Rigid docking is still a useful tool after 20 years
  - Flexible docking takes longer but allows the movement of protein residues
    - Usually mainly just the side-chains
    - Sometimes allows smaller movement of backbone
  - MD takes even longer to simulate ligand-binding event (except for local sampling only)
    - Allows the movement of everything in a selected environment within a selected amount of time
    - Accelerated methods available but still takes much longer than simple docking
- Assumptions:
  - with force-field based method assuming no reactions

Docking, redocking and cross docking

- Re-docking refers to the ability to reproduce co-crystallized binding geometry and orientation of the associated ligand given a rigid macromolecule state.
- Cross-docking refers to utilizing different ligand structures isolated from multiple PDB files of the same protein to test against a single rigid protein model structure.

Practical Example

Molecular Docking with AutoDock-related tools

Simple molecular graphics with UCSF Chimera

Recommended Programs

Program	Download Link
AutoDock4	http://autodock.scripps.edu/downloads
AutoDock VINA	http://vina.scripps.edu/download.html
AutoDock VINA 1.2 (Linux/Mac only)	https://github.com/ccsb-scripps/AutoDock-Vina/releases
AutoDockTools	https://ccsb.scripps.edu/mgltools/downloads (new) ~~http://mgltools.scripps.edu/downloads~~ (old) Note 1: Windows user need to set the path to the startup directory Note 2: MGLTools cannot run on Catalina OS
ADFR	https://ccsb.scripps.edu/mgltools/downloads (includes AGFR → easier setup for AG4/AD4)
VEGA ZZ (Windows only)	https://www.ddl.unimi.it/cms/index.php?Software_projects:VEGA_ZZ:Download
UCSF Chimera	https://www.cgl.ucsf.edu/chimera/download.html
Open Source PyMol	Instructions for Linux: https://pymolwiki.org/index.php/Linux_Install Instructions for Windows: https://pymolwiki.org/index.php/Windows_Install Instructions for Mac: https://pymolwiki.org/index.php/MAC_Install Via Anaconda: see https://anaconda.org/conda-forge/pymol-open-source
Avogadro	https://sourceforge.net/projects/avogadro/files/latest/download

Method 1 Easy but limited, Chimera-Vina interface

1. Download the protein file from the RCSB PDB
2. [Prepare ligand file]
3. Pre-processing with UCSF Chimera
   a. Select chain and save
   b. Model missing atoms from rotamer library
   c. Add hydrogens and save
   d. Save receptor file
   e. Save ligand file

Alternative methods can be used to predict ionizable residue's protonation based on pKa of the local environment at selected pH.

4. Using the Vina GUI in UCSF Chimera for docking
   a. Open receptor and ligand file
   b. Set output filename
   c. Select receptor
   d. Select ligand
   e. Setting up docking gridbox
   f. (remarks on restricted settings)
   g. Choosing the correct path to vina executable
   h. Run the program!
   i. [check the progress in the logfile]

Method 1 Easy but limited, Chimera-Vina interface

1. Download the protein file from the RCSB PDB
2. [Prepare ligand file]
3. Pre-processing with UCSF Chimera
4. Using the Vina GUI in UCSF Chimera for docking
5. Visualize docking results
   a. View the ranked binding pose and score
   b. Compare with crystal structure
   c. Detect hydrogen bonds
   d. Detect contacts

Method 1 Easy but limited, Chimera-Vina interface

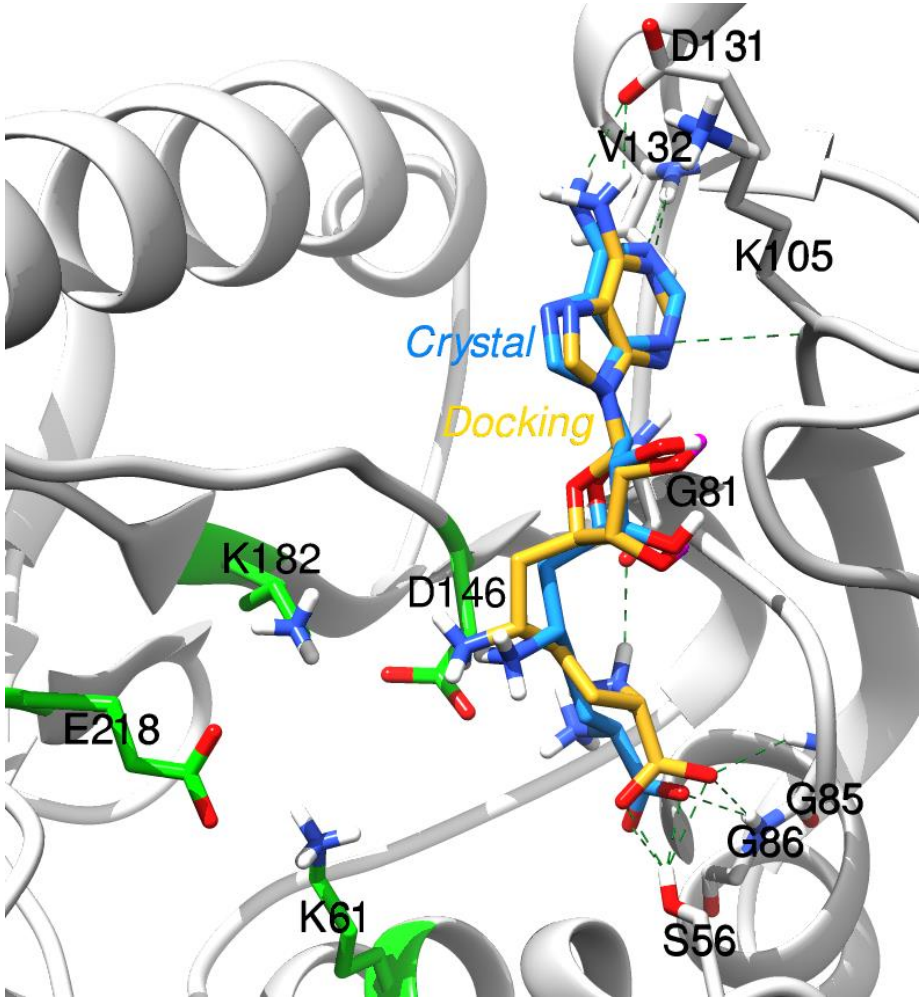
1. Download the protein file from the RCSB PDB
2. [Prepare ligand file]
3. Pre-processing with UCSF Chimera
4. Using the Vina GUI in UCSF Chimera for docking
5. Visualize docking results
6. Generate simple graphics for presentation
   a. Choose different colors for receptor and ligand(s)
   b. Highlight important residues if needed
   c. Changing pseudo-bond representation
   d. Placing residue labels
   e. Render the image (Chimera/POV-Ray)

Representation of ZIKV MTase-SFG complex

- Molecular graphics generated with UCSF Chimera and ray-traced with POV-Ray.
- Hydrogen bonds & distance detected by Chimera
- Residue labels and distance labels manually placed
- Protein is colored in dark grey except K-D-K-E catalytic tetrad highlighted in green
- Ligand is colored in dodger blue (crystal)/goldenrod (docking)
- Non-polar hydrogen atoms were hidden for clarity

In this system, using a rigid ligand can give a better score!

- Possible contribution of water-mediated interactions
- Large number of local minima on the energy surface with the flexible structure.



ZIKV MTase-Sinefungin
VINA score = -8.2 kcal/mol

Method 2a less convenient but more flexible, ADT-Vina interface

1. Prepare receptor and ligand according to step 1-3 of method 1
2. [windows only: set startup directory]
3. Load the receptor into AutoDockTools (ADT) and save as pdbqt
4. Prepare the ligand with ADT
   a. Load the ligand .pdb file
   b. Detect root
   c. Choose torsion: Use default setting as control
   d. Save the ligand prepared in (c) as .pdbqt to be the control
   e. Choose torsion: Set all rotatable bonds as rigid
   f. Save the ligand prepared in (e) as .pdbqt to test the setting
5. Setting up the gridbox with AGFR (Auto)
   a. Load the receptor .pdbqt
   b. With the gear button of AutoSite option, select AutoSite 1.1 and uncheck peptide scoring
   c. Run AutoSite 1.1
   d. Select the pocket you want and click “box all displayed ligand binding pockets”
   e. Uncheck “create gradient” box
   f. Click generate target file
   g. Quit
6. Unzip the target file generated by AGFR

You may also setup the docking box in ADT which is the “conventional” approach but its less efficient.

Method 2a less convenient but more flexible, ADT-Vina interface

1. Prepare receptor and ligand according to step 1-3 of method 1
2. [windows only: set startup directory]
3. Load the receptor into AutoDockTools (ADT) and save as pdbqt
4. Prepare the ligand with ADT
5. Setting up the gridbox with AGFR (Auto) or directly in ADT
6. Unzip the target file generated by AGFR
7. Go back to AutoDockTools to perform docking with **AutoDock Vina**
   a. Under “grid” open the .gpf from the AGFR-generated contents
   b. Check the gridbox
   c. Under “docking”:
      i. Open the “rigidReceptor.pdbqt” as the rigid filename
      ii. Open the ligand file of default setting
      iii. Save as Vina config, use default settings this time and check the filenames carefully
      iv. Repeat (ii)-(iii) for the rigid ligand
   d. Under “Run” select “Run AutoDock Vina”
      i. Select the correct path to the Vina executable
      ii. Select the vina config file of default settings
      iii. Run Vina!
      iv. Repeat (ii)-(iii) for the rigid ligand
8. View the docking results in UCSF Chimera (Tools > ViewDock) /AutoDockTools

Method 2b less convenient but more flexible, ADT-AD4 interface

1. Prepare receptor and ligand according to step 1-3 of method 1
2. [windows only: set startup directory]
3. Load the receptor into AutoDockTools (ADT) and save as pdbqt
4. Prepare the ligand with ADT
5. Setting up the gridbox with AGFR (Auto) or directly in ADT
6. Unzip the target file generated by AGFR
7. **Copy everything from the unzipped folder to the working directory that contains your ligand .pdbqt files**
8. Go back to AutoDockTools to perform docking with **AutoDock4**
   a. Under "grid" open the .gpf from the AGFR-generated contents
   b. Check the gridbox
   c. Under "docking":
      i. Open the "rigidReceptor.pdbqt" from Marcomolecule > set rigid filename
      ii. Open the ligand file of default setting from Ligand > open
      iii. In Search Parameters > Genetic Algorithm use "short" in Maximum no. of evals to be fast then click accept.
         - In the practical case better use long or more for accurate docking, or ok to further reduce the number in virtual screening.
      iv. Docking > output > Lamarckian GA and save the docking parameter .dpf file
   d. Under "Run" select "Run AutoDock4"
      i. Select the correct path to the AD4 executable
      ii. Select the .dpf file and check if the working directory is correct (especially for windows)
      iii. Launch AD4
9. View the docking results in AutoDockTools (Allowing more advanced analysis)
10. Export results as .pdbqt file to visualize in UCSF Chimera

If you adjust the gridbox in this step you'll need to rerun autogrid4 for AD4.

Next: try it yourself if you are interested!

- Including the water molecules in the active site?
- Changing docking exhaustiveness? (Vina)
- Changing GA settings? (AD4)

The command-line interface

- Utility scripts for doing everything without a GUI
- Sample bash script to prepare ligands with default torsions and run Vina

```
#!/bin/bash
ADTHOME="/opt/adt/MGLTools-1.5.6/MGLToolsPckgs/AutoDockTools"

for ligpdb in `cat list.txt`; do
    ligand=`basname ${ligpdb} .pdb`
    ${ADTHOME}/Utilities24/prepare_ligand4.py -l ${ligpdb}
    for protein in ZIKVMTase HCMTR1 HRNMT ; do
        if [[ ! -e ${protein}_docking ]] ; then
            mkdir ${protein}_docking/
        fi
        vina --center_x 19.390 --center_y 6.054 --center_z 3.936     \
            --size_x 15 --size_y 17.25 --size_z 14.25 \
            --cpu 8 --receptor ./${protein}.pdbqt --ligand ./${ligand}.pdbqt \
            --exhaustiveness 10 --num_modes 100 --energy_range 3.0 \
            --out ${protein}_docking/${ligand}.pdbqt --log ${protein}_docking/${ligand}.log
        ${ADTHOME}/Utilities24/pdbqt_to_pdb.py -f ${protein}_docking/${ligand}.pdbqt
    done
done
```

Prepare ligand file.

For each selected receptor, dock the ligand in it.

Convert the top-scoring docking pose of ligand into .pdb format.

Virtual screening with ADT cmd tools

- Automatic input -> docking with minimal scripting
- Possible to pipe to MD

Related command-line tools

- Updated version of Vina 1.2 that is only available for Linux/Mac
  - Improved functionality
  - Adding AD4 and Vinardo scoring
  - Additional docking protocols
  - Python3 support
- Smina
  - Fork of VINA that allows:
    - Customizable scoring function
    - Efficient setup of docking gridbox automatically based on a ligand structure
- QuickVina
  - Fork of VINA that:
    - Aims to speed up docking with comparable accuracy
- …

Reference

- Leach, A. *Molecular Modelling*; Pearson Education (US), 2001-01-15.
- AutoDock4 + ADT tutorial
  - http://autodock.scripps.edu/faqs-help/tutorial/using-autodock-4-with-autodocktools
- More updated & practical ADT tutorials from Bevan & Brown's Lab
  - https://osf.io/82n73/wiki/Docking%2C%20Re-docking%2C%20and%20Cross%20Docking/
- Even more updated Vina tutorial using the Unix-only Vina 1.2
  - https://autodock-vina.readthedocs.io/en/latest/docking_requirements.html